



Project 3: Hadoop and Spark

[Project 3: Hadoop and Spark](#)[Grading Rubric](#)[Resources & AI Policy](#)[LLM Usage Policy](#)

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Due: 11:59 PM, April 23

The goal of this project is to deepen your understanding of data analytics stacks through practical, hands-on experience. You will set up a data analytics environment using Apache Hadoop for distributed storage (HDFS) and Apache Spark as the computation engine. You will then develop and execute several applications to demonstrate your skills.

Tasks:

1. **Cluster Setup:** Launch a cluster of virtual machines (VMs) in a cloud environment (e.g., AWS, Azure, GCP). Your setup should include one master node and at least two worker (slave) nodes.
2. **Deploy HDFS:** Install and configure the Hadoop Distributed File System (HDFS) across your cluster.
3. **Data Preparation:** Download the text version of Pride and Prejudice from Project Gutenberg and store this file on your HDFS cluster.
4. **Deploy Apache Spark:** Set up Apache Spark on your Hadoop cluster, ensuring it can interact with the HDFS.
5. **Word Count with Spark:** Using Spark, write and execute a word count program that processes the stored file in HDFS. Your program should count occurrences of each word, sort the words by frequency in descending order, and output the top 20 most frequent words along with their counts.
6. **Estimating π with Monte Carlo Method in Spark:** Implement a Spark application using the Monte Carlo method to estimate the value of π .

Since the area of a circle with radius r is given by $A = \pi r^2$, the area of the unit circle is π . Using a Monte Carlo simulation, we can estimate the value of π by randomly sampling points within a square and counting the number of points that fall within the unit circle.

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square and determine the proportion of points that fall within the unit circle. Multiplying this proportion by 4 (the area of the square) yields an estimate of π .

7. **[Graduate Students Only] Performance Benchmarking & Scalability Analysis:** Conduct a systematic performance study of your Spark cluster. Re-run the Word Count job under at least three different configurations (e.g., varying the number of worker nodes, executor memory, or degree of parallelism). Record wall-clock execution times and resource utilization for each configuration. Analyze your results: do they exhibit linear speedup? Where are the bottlenecks? Discuss your findings in relation to theoretical models of parallel computation (e.g., Amdahl's Law or Gustafson's Law).

Submission Guidelines:

Submit a comprehensive report as a PDF to Brightspace that includes:

- The commands executed for each task.
- Source code for all Spark applications.
- Detailed observations and outputs for each step.
- Clear explanations outlining the purpose and rationale behind each task performed.

Your report should clearly demonstrate both your technical skills and conceptual understanding of the Hadoop and Spark ecosystems.

Here is the updated grading rubric and LLM usage policy tailored specifically for your Hadoop and Spark project.

Grading Rubric

Criterion	Description	Points
Cloud & HDFS Infrastructure	Successful provisioning of the cloud cluster (Master + 2 Workers) and correct configuration of HDFS. Evidence of HDFS daemons (NameNode, DataNodes) running and communicating across the network.	30
Spark Deployment & Integration	Functional Apache Spark installation. Demonstration that Spark is correctly co	20

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Criterion	Description	Points
	with the HDFS layer (YARN or Standalone mode) and can submit jobs to the cluster.	
Word Count Application	Correct implementation of the Word Count logic (counting, sorting by frequency, limiting to top 20). Successful ingestion of <i>Pride and Prejudice</i> from HDFS and generation of correct output.	20
Monte Carlo Pi Simulation	Correct mathematical implementation of the Monte Carlo method. Application successfully parallelizes the workload across the Spark cluster to estimate π .	20
Report, Analysis & Quality	Clear presentation of commands, logs, and outputs. Code is well-organized. Detailed observations and rationale provided for the setup and execution steps.	10
[Graduate Only] Performance Benchmarking & Scalability Analysis	Systematic experimentation across at least three configurations, correct measurement of execution time and resource utilization, and a substantive discussion connecting empirical results to theoretical models (e.g., Amdahl's Law, Gustafson's Law).	20
Total (Undergraduate)		100
Total (Graduate)	Graded out of 120, then normalized to 100.	120 → 100

Undergraduate Bonus: Undergraduate students who complete Task 7 (the graduate-only task) will receive a **10% bonus** added to their final project score, for a maximum possible score of 110.

Resources & AI Policy

You are expected to primarily rely on official online documentation to research and understand how to complete each project task. Key resources include:

- **Apache Hadoop Documentation:** <https://hadoop.apache.org/docs/current/hadoop-project-dist/hadoop-common/quickstart.html>
- **Apache Spark Documentation:** <https://spark.apache.org/docs/latest/quickstart.html>

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- **Cloud Provider Docs:** Documentation for your chosen provider (AWS EC2, Google Compute Engine, or Azure VMs).

LLM Usage Policy

You are **ALLOWED** to use LLM-based AI assistants (e.g., ChatGPT, Claude, GitHub Copilot) as a supplementary tool. These can be helpful for:

- Generating boilerplate code (e.g., PySpark or Scala shells).
- Debugging specific Java/Python stack traces or configuration errors (e.g., `core-site.xml` issues).
- Explaining the parameters of specific Spark transformations and actions.

However, the following strict guidelines apply:

1. **Validation is Mandatory:** You are fully responsible for validating the correctness, security, and applicability of any information or code provided by an LLM. Directly submitting unverified or misunderstood LLM output is not acceptable.
2. **Context Matters:** LLMs often assume “local” setups. You must ensure any generated configuration matches your specific distributed cluster environment (IP addresses, ports, memory limits).
3. **Accountability:** You must be able to explain every line of code and every configuration change in your final submission. If asked, you should be able to walk through how your Word Count sort logic works without relying on the AI.

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